

SECTION A*(25 marks)***1.** Evaluate

(a) $\lim_{x \rightarrow 3} \frac{\sqrt{x-1} - \sqrt{2}}{9 - x^2}.$ **(3 marks)**

(b) $\lim_{x \rightarrow -\infty} \frac{4x + \sqrt{x^2 - 3}}{x}.$ **(3 marks)**

(c) $\lim_{x \rightarrow 4^+} \frac{16 - x^2}{|x - 4|}.$ **(3 marks)**

2. Find the derivatives of the following functions

(a) $y = \ln(x^2 \sqrt{x-1}).$ **(4 marks)**

(b) $y = 4 \sin\left(\frac{e^x}{\ln x}\right).$ **(4 marks)**

3. A function f is defined by $f(x) = 3x^2 - x^3.$

(a) Find the coordinates of the stationary points and determine the nature of the points. **(3 marks)**

(b) Given that the point (1,2) lies on the curve, show that it is the point of inflection. Hence, sketch the graph of $f(x).$ **(5 marks)**

SECTION B

(75 marks)

1. Given a complex number $z = \frac{(1-i)^2}{3+i}$.
- (a) Express z in Cartesian form $a+bi$, where a and b are real numbers. (2 marks)
- (b) Hence, find the modulus and argument of z . (3 marks)
2. Determine the solution set of the following inequalities
- (a) $\frac{1}{3-x} \geq \frac{2}{x-7}$. (6 marks)
- (b) $\left| \frac{x}{x-4} \right| < 1$. (7 marks)
3. Given that $f(x) = e^{3x} + 2$, $x \in \mathbb{R}$.
- (a) Find $f^{-1}(x)$. (3 marks)
- (b) Sketch the graphs of f and f^{-1} on the same axes.
Hence, state the domain and range of f and f^{-1} . (6 marks)
4. (a) Given $f(x) = \ln(2x-1)$ and $g(x) = 2e^{-x} + 1$.
Find $g \circ f(x)$, hence evaluate $g \circ f(4)$. (6 marks)
- (b) Find the domain and range of $f(x) = 3\sin 2x$, $0 \leq x \leq 2\pi$,
Hence, sketch the graph of $f(x)$. (3 marks)

5. (a) (i) If g is a function with $g'(2) = 3$, find $\lim_{x \rightarrow 2} \frac{g(x) - g(2)}{\sqrt{2x} - 2}$. (3 marks)

- (ii) Evaluate $\lim_{x \rightarrow 2} \frac{1 + x^2}{(x - 2)(x + 1)}$. (3 marks)

- (b) A function f is defined as follows

$$f(x) = \begin{cases} x^3 - 3 & , \quad x \leq -1 \\ a + (x + 3)^2 & , \quad -1 < x \leq 3 \\ bx + 4 & , \quad 3 < x \leq 6 \\ 8 & , \quad x > 6 \end{cases}$$

- (i) If f is continuous at $x = -1$ and $x = 3$, find the values of a and b . (5 marks)

- (ii) Determine the values of b such that f is not continuous at $x = 6$. (3 marks)

6. (a) Given the parametric equation

$$x = 4t + \frac{3}{t}, \quad y = 4t - \frac{3}{t} \quad \text{where } t \neq 0.$$

- (i) Show that $\frac{dy}{dx} = 1 + \frac{6}{4t^2 - 3}$. (5 marks)

- (ii) Find the value of $\frac{d^2y}{dx^2}$ when $t = 1$. (4 marks)

- (b) Given that $f(x) = \frac{1}{\sqrt{5x + 36}}$.

Show that $\frac{f''(x) + f'(x)}{\sqrt{5x + 36}} = \frac{75}{4} [f(x)]^6 - \frac{5}{2} [f(x)]^4$. (6 marks)

7. Water is pumped at a constant rate of $180 \text{ cm}^3\text{s}^{-1}$ into an empty inverted conical container of radius 30 cm and height 75 cm.
- (a) Show that the rate of change of the height of water in the container is $\frac{dh}{dt} = \frac{1125}{\pi h^2}$, where h is the height of water in centimeters at time, t , seconds. **(5 marks)**
- (b) Calculate the rate of change of the radius of water in the container at the instant the radius of the surface of water is one third of the radius of the container. Give your answer in terms of π . **(3 marks)**
- (c) Find the time taken in minutes to fill the container completely. Give your answer in 3 significant figures. **(2 marks)**

END OF QUESTIONS PAPER

ANSWER

SECTION A

1. (a) $-\frac{1}{12\sqrt{2}}$

(b) 3

(c) -8

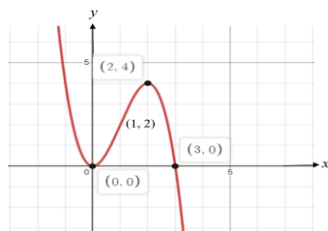
2. (a) $\frac{5x-4}{2x(x-1)}$

(b) $\frac{dy}{dx} = 4 \cos\left(\frac{e^x}{\ln x}\right) \cdot \frac{xe^x \ln x - e^x}{x(\ln x)^2}$

3. (a) Stationary points are (0, 0) and (2, 4)

Minimum point : (0, 0)

Maximum point : (2, 4)



SECTION B

1. (a) $z = -\frac{1}{5} - \frac{3}{5}i$

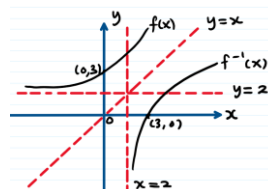
(b) $|z| = \frac{\sqrt{10}}{5}$ or 0.6325
 $\text{Arg}(z) = -1.8925$ rad

2. (a) Solution set : $\left\{x : x < 3 \text{ or } \frac{13}{3} \leq x < 7\right\}$

(b) Solution set : $\{x : x < 2\}$

3. (a) $f^{-1}(x) = \frac{1}{3} \ln(x-2)$

(b)



$D_f = (-\infty, \infty)$

$R_f = (2, \infty)$

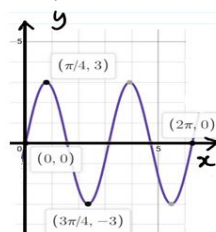
$D_{f^{-1}} = (2, \infty)$

$R_{f^{-1}} = (-\infty, \infty)$

4. (a) $g(f(x)) = \frac{2x+1}{2x-1}, x \neq \frac{1}{2}$

$g(f(4)) = \frac{9}{7}$

(b) $D_f = [0, 2\pi], R_f = [-3, 3]$



5. (a) (i) 6

(ii) $\lim_{x \rightarrow 2} \frac{1+x^2}{(x-2)(x+1)}$ does not exist

(b) (i) $a = -8, b = 8$

(ii) $b \neq \frac{2}{3}$

6. (a)(ii) -48

7. (b) $\frac{dr}{dt} = \frac{18}{25\pi}$ cm/s

(c) $t = 6.54$ minutes

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